

Kinetic Energy, Work-Energy Theory and Conservative Force

SS ILP Physics (Intermediate), Week 3 Class 4

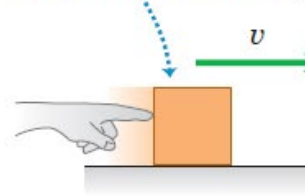


Outline

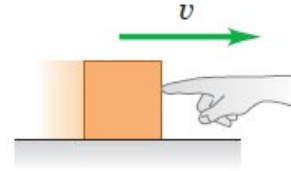
- Kinetic energy
- Work-energy theorem
- Conservative Force

Dynamics of an Object and Work Done

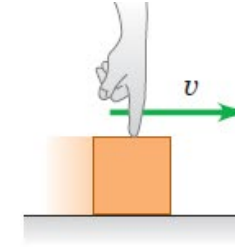
A block slides to the right on a frictionless surface.



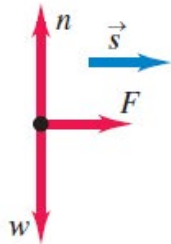
If you push to the right on the moving block, the net force on the block is to the right.



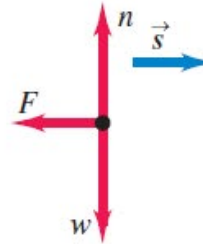
If you push to the left on the moving block, the net force on the block is to the left.



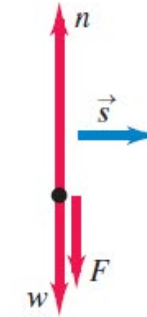
If you push straight down on the moving block, the net force on the block is zero.



- The total work done on the block during a displacement $\vec{s}$ is positive: $W_{\text{tot}} > 0$.
- The block speeds up.



- The total work done on the block during a displacement $\vec{s}$ is negative: $W_{\text{tot}} < 0$.
- The block slows down.



- The total work done on the block during a displacement $\vec{s}$ is zero: $W_{\text{tot}} = 0$.
- The block's speed stays the same.

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Kinetic Energy: Definition

- A particle is accelerated by a force F directed along the positive x direction. Suppose the speed changes from v_1 to v_2 while the particle makes a displacement of $s = x_2 - x_1$

- Using kinematics of linear motion:

$$v_2^2 = v_1^2 + 2as$$

$$a_x = \frac{v_2^2 - v_1^2}{2s}$$

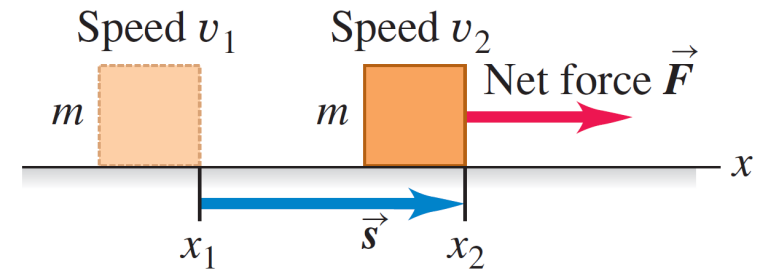
- From Newton's 2nd Law

$$F = ma_x = m \frac{v_2^2 - v_1^2}{2s}$$

$$Fs = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

- The quantity Fs is the **total work done applied to the particle**, W_{tot} .

- Define the **kinetic energy (KE)** of the particle as $K = \frac{1}{2}mv^2$



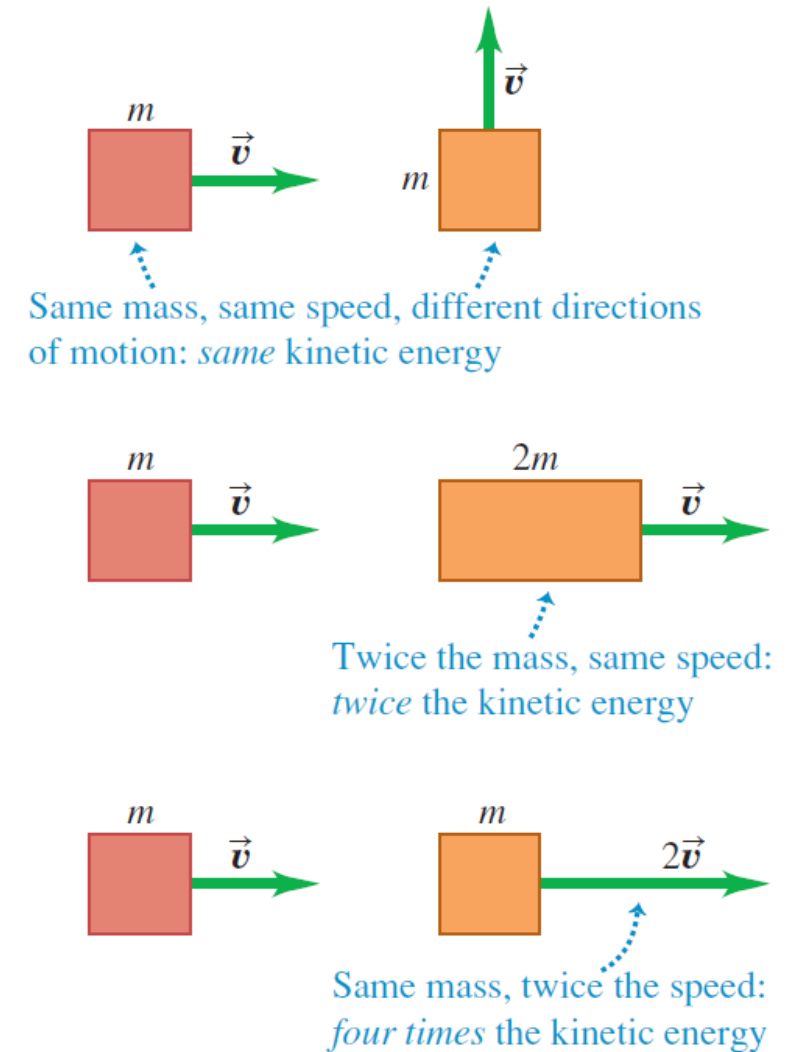
H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Kinetic Energy: A Closer Look

- Kinetic energy of a particle moving with speed v :

$$K = \frac{1}{2}mv^2$$

- KE is a scalar quantity – depends only on the mass and the speed of the particle
- KE is always greater than zero
- KE is zero when the particle is at rest
- SI unit of KE: Joule (J)



H.D. Young and R. A. Freedman, Sears and Zemansky's
University Physics. 13E, Addison-Wesley, Boston, 2012.

Example

Adult cheetahs, the fastest of the great cats, have a mass of about 70 kg and have been clocked running at up to 32 m/s . (a) How many joules of kinetic energy does such a swift cheetah have? (b) By what factor would its kinetic energy change if its speed were doubled?

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Example: Solution

(a) Kinetic energy: $K = \frac{1}{2}mv^2 = \frac{1}{2} \cdot 70 \text{ kg} \cdot (32 \text{ ms}^{-1})^2 = 36,000 \text{ J}$

(b) If the speed is doubled, the kinetic energy becomes:

$$K' = \frac{1}{2}m(2v)^2 = 4 \cdot \frac{1}{2}mv^2 = 4K$$

The kinetic energy would be increased by a factor of 4 if the speed is doubled.

Concept Question

A truck moving at 15 m/s has a kinetic energy of $4.2 \times 10^5 \text{ J}$. By what multiplicative factor does the kinetic energy of the truck increase if the speed is tripled?

- a) By a factor of 1.
- b) By a factor of 3.
- c) By a factor of 4.
- d) By a factor of 9.

Concept Question: Solution

A truck moving at 15 m/s has a kinetic energy of $4.2 \times 10^5 \text{ J}$. By what multiplicative factor does the kinetic energy of the truck increase if the speed is tripled?

- a) By a factor of 1.
- b) By a factor of 3.
- c) By a factor of 4.
- d) **By a factor of 9.**

If the speed is tripled, the kinetic energy becomes:

$$K' = \frac{1}{2}m(3v)^2 = 9 \cdot \frac{1}{2}mv^2 = 9K$$

Case Problem 1

- a) In the Bohr model of the atom, the ground-state electron in hydrogen has an orbital speed of 2190 km/s. What is its kinetic energy?
[Note: mass of an electron is 9.11×10^{-31} kg]
- b) If you drop a 1.0-kg weight from a height of 1.0 m, how many joules of kinetic energy will it have when it reaches the ground?
- c) Is it reasonable that a 30-kg child could run fast enough to have 100 J of kinetic energy?

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Case Problem 1: Solution

(a) Kinetic energy of a ground-state electron:

$$K_{gs} = \frac{1}{2}mv^2 = \frac{1}{2} \cdot 9.11 \times 10^{-31} \text{ kg} \cdot (2190 \times 1000 \text{ ms}^{-1})^2 = 2.18 \times 10^{-18} \text{ J}$$

(b) From linear kinematics,

$$\begin{aligned} v_f^2 &= v_0^2 + 2gs \\ v_f &= \sqrt{v_0^2 + 2gs} = \sqrt{2gs} \\ K &= \frac{1}{2}mv_f^2 = \frac{1}{2}m \cdot 2gs = mgs = 1.0 \text{ kg} \times 9.8 \text{ ms}^{-2} \cdot 1.0 \text{ m} = 9.8 \text{ J} \end{aligned}$$

(c) Kinetic energy is related to speed by

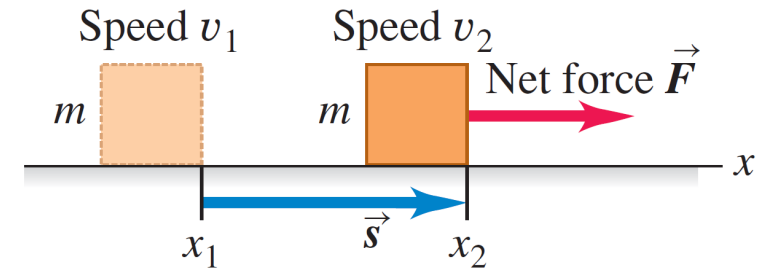
$$\begin{aligned} K &= \frac{1}{2}mv^2 \\ v &= \sqrt{\frac{2K}{m}} = \sqrt{2 \times \frac{100}{30}} \text{ ms}^{-1} = 2.6 \text{ ms}^{-1} \end{aligned}$$

Work-Energy Theorem

- From Newton's 2nd Law

$$Fs = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$Fs = K_2 - K_1$$



- The quantity Fs is the **total work done applied to the particle**, W_{tot} .
- Work-Energy Theorem:** The work done by the net force on a particle equals the change in the particle's kinetic energy

$$W_{tot} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = K_2 - K_1 = \Delta K$$

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Work-Energy Theorem and Kinetic Energy

- Work-energy theorem:

$$\boxed{W_{tot} = K_2 - K_1 = \Delta K}$$

- $W_{tot} > 0$, the KE of the particle **increases** (final KE > initial KE, i.e. $K_2 > K_1$).
- $W_{tot} < 0$, the KE of the particle **decreases** (final KE < initial KE, i.e. $K_2 < K_1$).
- Example: A particle initially at rest, accelerate to a final velocity of v by W_{tot} :

$$W_{tot} = \frac{1}{2}mv^2 - 0 = \frac{1}{2}mv^2$$

- The KE of a particle is equal to the total work that was done to accelerate it from rest to its present speed, v .

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Concept Question

How much work is required for a 74-kg sprinter to accelerate from rest to 2.2 m/s?

- a) 160 J
- b) 180 J
- c) 320 J
- d) 360 J

Concept Question: Solution

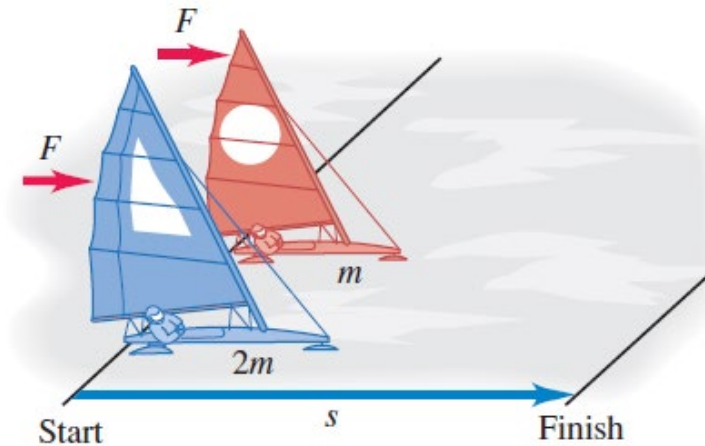
How much work is required for a 74-kg sprinter to accelerate from rest to 2.2 m/s?

- a) 160 J
- b) 180 J
- c) 320 J
- d) 360 J

$$W = \Delta K = K_{final} - 0 = \frac{1}{2} (74 \text{ kg}) \left(2.2 \frac{\text{m}}{\text{s}} \right)^2 = 180 \text{ J}$$

Example

Two iceboats like hold a race on a frictionless horizontal lake. The two iceboats have masses m and $2m$. The iceboats have identical sails, so the wind exerts the same constant force, $\vec{F}$, on each iceboat. They start from rest and cross the finish line a distance s away. Which iceboat crosses the finish line with greater kinetic energy?



H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Example: Solution

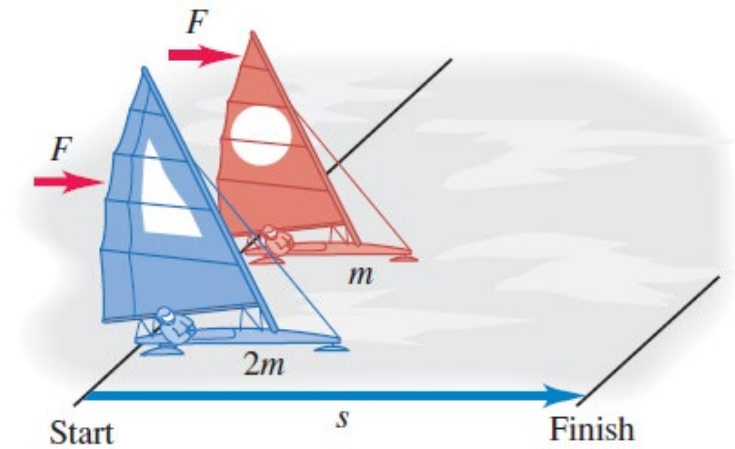
From work-energy theorem,

$$W = \Delta K = K_{final} - 0$$

$$K_{final} = \vec{F} \cdot \vec{s}$$

The final kinetic energy depends only on the dot product $\vec{F} \cdot \vec{s}$, and does not depend on mass. Hence, both iceboats will arrive the finishing line with **the same kinetic energy**.

Think: Which iceboat arrive at the finishing line with a faster velocity?



H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Case Problem 2

You throw a 0.145-kg baseball straight up. Suppose your hand moves upward by 0.50 m while you are throwing the ball. The ball leaves your hand with an upward velocity of 20.0 m/s (a) Find the magnitude of the force (assumed constant) that your hand exerts on the ball. (b) Find the speed of the ball at a point 15.0 m above the point where it leaves your hand. Ignore air resistance.

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Case Problem 2: Solution

- (a) Work done by the force exerted on the ball by your hand when moving the ball through a distance of $s_{12} = 0.50 \text{ m}$ is $W_F = F s_{12}$.

Work done by the gravity force on the ball is:

$$W_{grav} = -mgs_{12}$$

Hence, the net work done on the ball is

$$W_{net} = W_F + W_{grav} = (F - mg)s_{12}$$

From work-energy theorem,

$$W_{net} = \Delta K = K_2 - K_1 = \frac{1}{2}mv_2^2 - 0$$

$$\text{Hence, } (F - mg)s_{12} = \frac{1}{2}mv_2^2$$

$$F = m \left(\frac{v_2^2}{2s_{12}} + g \right) = 59.4 \text{ N}$$

Case Problem 2: Solution

(b) Find $W_{23} = -mgh$ where $h = 15 \text{ m}$. Work-energy theorem gives

$$\begin{aligned} W_{23} &= \Delta K_{23} \\ -mgh &= K_3 - K_2 \\ -mgh &= \frac{1}{2}mv_3^2 - \frac{1}{2}mv_2^2 \end{aligned}$$

$$v_3 = \sqrt{2\left(\frac{v_2^2}{2} - gh\right)} = \sqrt{2\left(\frac{20^2}{2} - 9.8 \times 15\right)} \text{ ms}^{-1} = 10.3 \text{ ms}^{-1}$$

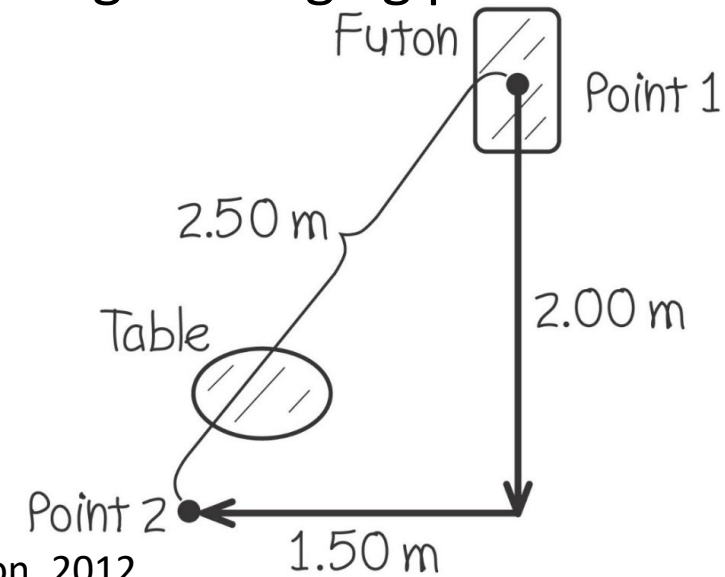
What is the meaning of work done?

- **Work done**, W , on a system by a constant force on the system is the product of the magnitude F of the force, the magnitude Δr of the displacement of the point of application of the force, and $\cos \theta$, where θ is the angle between the force and the displacement vectors (ie resolving displacement in the direction of the force).
- $W = F \Delta r \cos \theta = \vec{F} \cdot \Delta \vec{r}$

Case Problem 3: Work Done by Frictional Force

You are rearranging your furniture and wish to move a 40.0-kg futon 2.50 m across the room. A heavy coffee table, which you don't want to move, blocks this straight-line path. Instead, you slide the futon along a dogleg path; the doglegs are 2.00 m and 1.50 m long. The coefficient of kinetic friction is $\mu_k = 0.20$.

- Find the work done by frictional force when pushing the futon along the straight-line path connecting points 1 and 2.
- Find the work done by frictional force when pushing the futon along the dogleg path.



H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

Case Problem 3: Work Done by Frictional Force

Frictional (kinetic) force between the futon and the floor is given by $F_k = \mu_k N = \mu_k mg$. (negative means frictional force is pointing opposite to the direction of motion that you push the item).

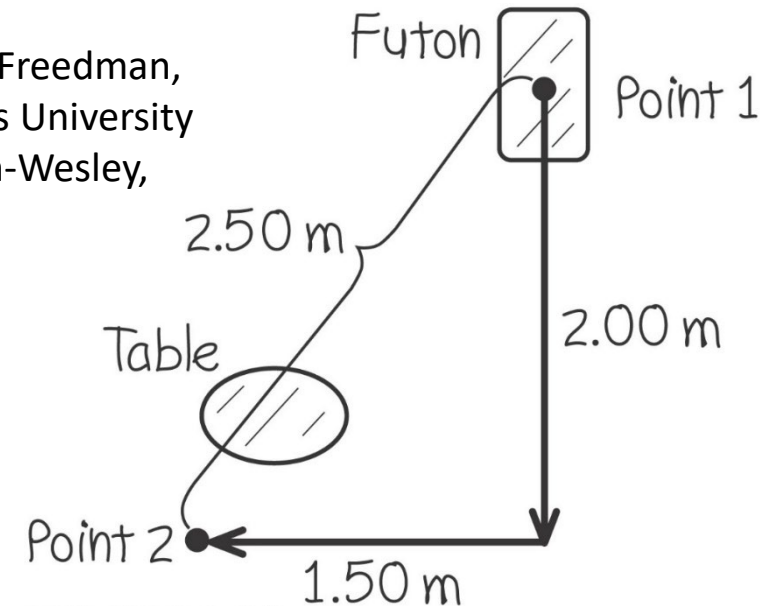
Work done along the straight path,

$$\begin{aligned} W_I &= -f_k d = -\mu_k mgd \\ &= -0.20 \times 40.0 \text{ kg} \times 9.81 \text{ ms}^{-2} \times 2.50 \text{ m} \\ &= -196 \text{ J} \end{aligned}$$

Work done along the dogleg path ,

$$\begin{aligned} W_{II} &= -f_k d = -\mu_k mg(d_1 + d_2) \\ &= -0.20 \times 40.0 \text{ kg} \times 9.81 \text{ ms}^{-2} \times (2.00 \text{ m} + 1.50 \text{ m}) \\ &= -274 \text{ J} \end{aligned}$$

H.D. Young and R. A. Freedman,
Sears and Zemansky's University
Physics. 13E, Addison-Wesley,
Boston, 2012.

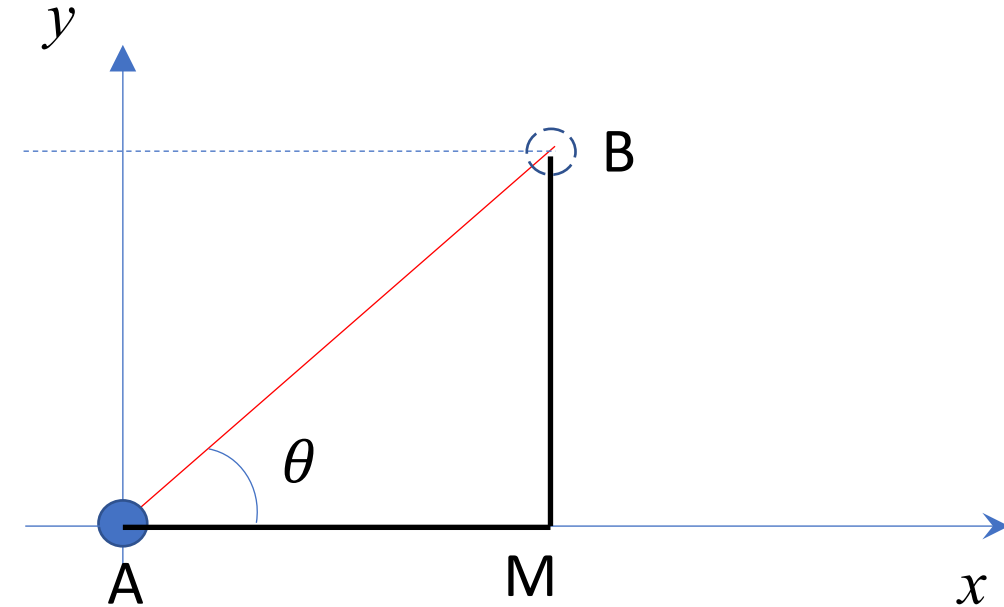


Frictional work depends on the path and you need additional $\delta W = 78 \text{ J}$ for dogleg compared to straight path.

Example: Work Done by Gravitational Force

Calculate the work done by $\vec{W} = mg(-\hat{j})$ on the object moving

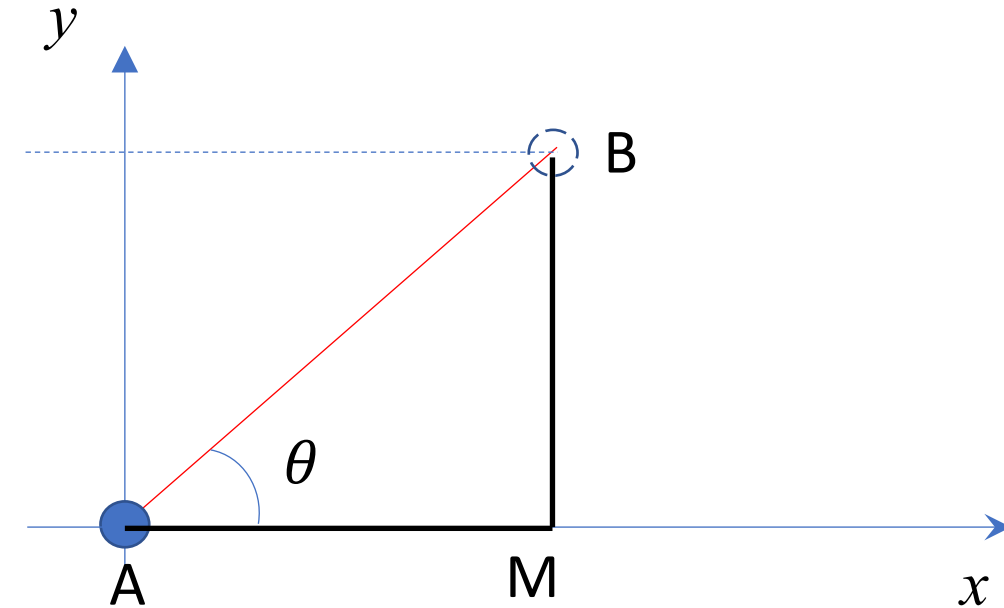
- a) along the path from A to B via point M (A to M then to B), and
- b) along the straight-line path connecting A to B.



Example: Solution

a) Work done along the path from A to M then to B,

$$\begin{aligned} W_{\text{I}} &= W_{\text{I},1} + W_{\text{I},2} = \vec{W} \cdot \Delta\vec{r}_1 + \vec{W} \cdot \Delta\vec{r}_2 \\ &= mg(-\hat{j}) \cdot |AM|\hat{i} + mg(-\hat{j}) \cdot |BM|\hat{j} = -mg|BM| \end{aligned}$$



b) Work done along the straight-line path,

$$\begin{aligned} W_{\text{II}} &= \vec{W} \cdot \Delta\vec{r} = mg(-\hat{j}) \cdot (|AB| \cos \theta \hat{i} + |AB| \sin \theta \hat{j}) \\ &= -mg|AB| \sin \theta = -mg|BM| \end{aligned}$$

Conservative Force

- There exist certain forces possessing a special feature that the work done by it is dependent only on initial and final positions of a motion of but not on the details of path followed (see Example).
- This type of forces are **conservative forces**.
- A force (such as friction, air drag) that is not conservative is called a ***non-conservative force*** (Case problem 3).

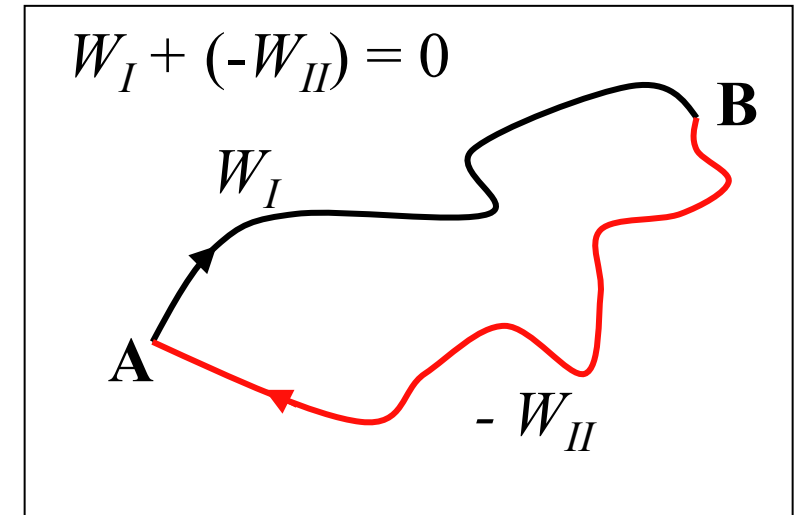
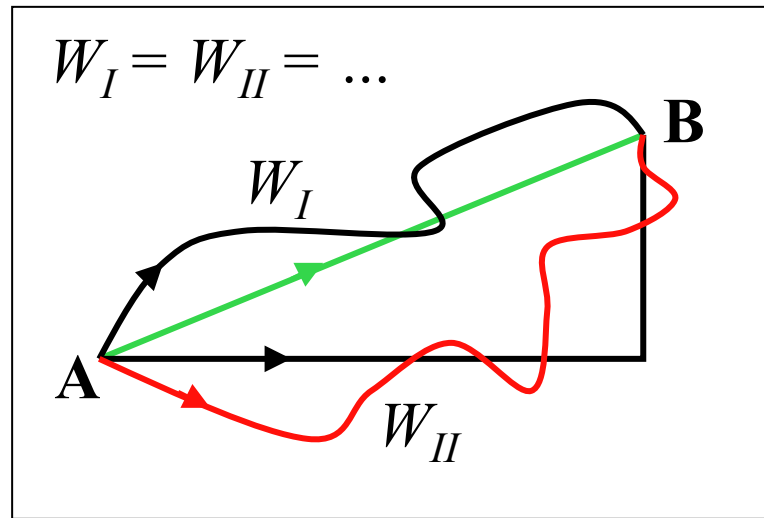
R.A. Serway and J.W. Jewett, Physics for Scientists and Engineers with Modern Physics. 9E, Cengage Learning, Boston, 2014.

Properties of Conservative Forces

- (Property I) A force is **conservative** if the work it does on a particle moving between any two points is **independent** of the path taken by the particle.



- (Property II) Work done by a conservative force exerted on a particle moving through any **closed path** is **zero**.



Off-class practice question(s)

Concept Question

A truck moving at 15 m/s has a kinetic energy of $4.2 \times 10^5 \text{ J}$. Find its mass.

- a) 3500 kg
- b) 3700 kg
- c) 7000 kg
- d) 7400 kg

Concept Question: Solution

A truck moving at 15 m/s has a kinetic energy of $4.2 \times 10^5 \text{ J}$. Find its mass.

- a) 3500 kg
- b) 3700 kg
- c) 7000 kg
- d) 7400 kg

$$K = \frac{1}{2}mv^2 \rightarrow m = \frac{2K}{v^2} = 3700 \text{ kg}$$

Case Problem

About 50,000 years ago, a meteor crashed into the earth near present-day Flagstaff, Arizona. Measurements from 2005 estimate that this meteor had a mass of about 1.4×10^8 kg and hit the ground at a speed of 12 km/s.

- (a) How much kinetic energy did this meteor deliver to the ground?
- (b) How does this energy compare to the energy released by a 1.0-megaton nuclear bomb?

Note: A megaton bomb releases the same amount of energy as a million tons of TNT, and 1.0 ton of TNT releases 4.184×10^9 J of energy.

Case Problem: Solution

(a) The kinetic energy of the meteor upon hitting the ground is

$$K = \frac{1}{2}mv^2 = \frac{1}{2} \times 1.4 \times 10^8 \text{ kg} \times (12 \times 1000 \text{ ms}^{-1})^2 = 1.0 \times 10^{16} \text{ J}$$

(b) Energy released by 1.0 megaton nuclear bomb is

$$E_{nuke} = 10^6 \times 4.184 \times 10^9 \text{ J} = 4.184 \times 10^{15} \text{ J}$$

Comparing the kinetic energy of the meteor and 1.0 megaton nuclear bomb:

$$N_{nuke} = \frac{K}{E_{nuke}} = 2.4$$

The kinetic energy of the meteor is equivalent to 2.4 one-megaton nuclear bomb.